

Homework 3

ASEN 5335 Aerospace Environment

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Problem 1: Plotting the Ionosphere

(i.)

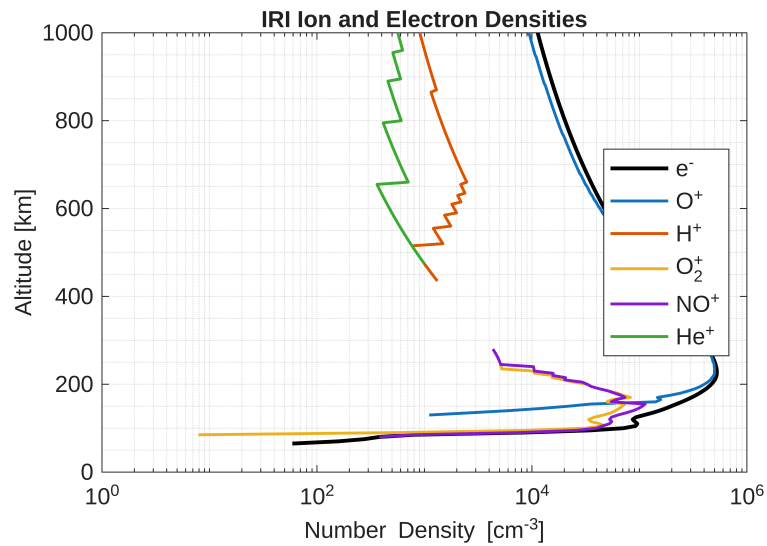


Figure 1: IRI Output

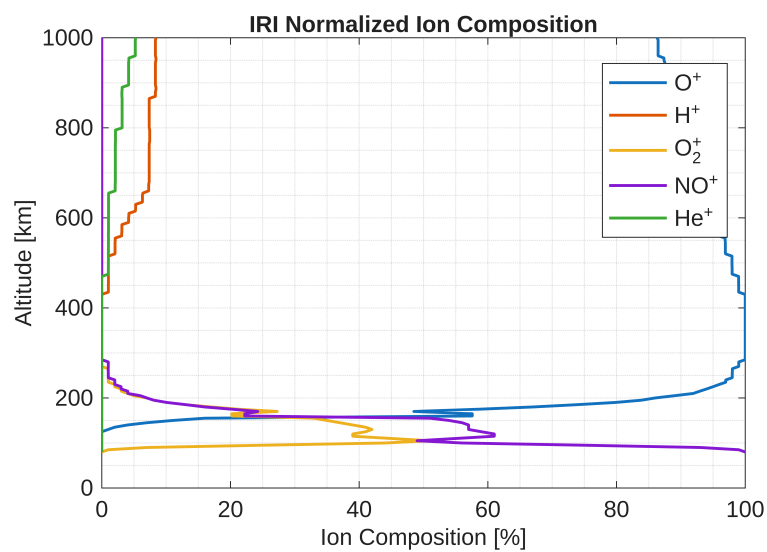


Figure 2: IRI Output Normalized

(ii.)

Below 500 km, we can observe that NO^+ dominates between 0-160 km. O_2^+ contributes more than 20% at between 90 and 175 km.

(iii.)

By convention 1 TECU is 10^{16} el/m². Integrating the total electrons from the IRI output, we find the TEC of the entire IRI output to be 12.47 TECU. When the peak electron density is 525696 m^{-3} , we can integrate the topside and bottomside separately to find they are 3.55 TECU and 8.92 TECU respectively. Normalizing against the total TEC found, we find they are 28.49% and 71.51% respectively.

Problem 2: Peak Ionization

(i.)

E Region

We are able to use the Chapman relation equation,

$$n_e(z, \theta) = \left(\frac{q_{max}}{\alpha_{eff}} \right)^{1/2} \exp \left[\frac{1}{2} (1 - z_1 - e^{-z_1}) \right]$$

Solving for q_{max} ,

$$q_{max} = \alpha_{eff} n_e^2 \exp[-(1 - z_1 - e^{-z_1})]$$

At the peak altitude $z = z_{max}$, $z_1 = 0$, so the exponential term is 1 and $n_e = n_{e,max}$:

$$q_{max} = \alpha_{eff} n_{e,max}^2$$

$$q_{max_E} = (10^{-8})(1.5 \times 10^5)^2$$

$$\boxed{q_{max_E} = 225 \text{ cm}^2/\text{s}}$$

F1 Region

Repeat the calculation above for the F1 Region,

$$q_{max_{F1}} = (7 * 10^{-9})(2.5 \times 10^5)^2$$

$$\boxed{q_{max_E} = 437.5 \text{ cm}^2/\text{s}}$$

F2 Region

Repeat the calculation above for the F2 Region,

$$q_{max_{F2}} = (5 * 10^{-9})(5.0 \times 10^5)^2$$

$$\boxed{q_{max_{F2}} = 1250 \text{ cm}^2/\text{s}}$$

(ii.)

The reason why the ionization peak altitude z_{max} moves upwards with increasing solar zenith angle χ when all other factors are kept constant is because when examining the equation,

Problem 3: Spacecraft Charging in LEO

We are able to use the charging equation,

$$\frac{dQ}{dt} = I_e + I_i + I_{SEE,i} + I_b + I_{ph}$$

By assuming $\frac{dQ}{dt} = 0$, the eclipse term, $I_{ph} = 0$, and no secondary electron emission or backscattered electron current, $I_{SEE,i} = I_b = 0$, the equation simplifies to,

$$0 = I_e + I_i$$

Using equations for I_i , and I_e ,

$$I_i = qn_i v_0 A_i$$

$$I_e = -\frac{1}{4} q n_e v_{e,th} A_e e^{qV/kT_e}$$

and calculating

$$V_{fl} = \frac{k_B T_e}{q} \ln \left(\frac{4v_0 A_i}{v_{e,th} A_e} \right)$$

The thermal voltage at a constant $T_e = 1000$ K is,

$$\frac{k_B T_e}{q} = \frac{(1.381 \times 10^{-23} \text{ J/K})(1000 \text{ K})}{1.602 \times 10^{-19} \text{ K}} = 0.0862 \text{ V}$$

The orbital velocity at 400 km altitude is,

$$v_0 = \sqrt{\frac{\mu}{R_E + h}} = \sqrt{\frac{3.986 \times 10^{14} \text{ m}^3/\text{s}^2}{6.378 \times 10^6 \text{ m} + 4.0 \times 10^5 \text{ m}}} = 7.67 \text{ km/s}$$

The electron thermal velocity is,

$$v_{e,th} = \sqrt{\frac{3k_B T_e}{m_e}} = \sqrt{\frac{3(1.381 \times 10^{-23} \text{ J/K})(1000 \text{ K})}{9.109 \times 10^{-31} \text{ kg}}} = 213 \text{ km/s}$$

If this is a 6U Cubesat, ions are collected only on the 20×30 cm ram face, electrons collect over entire surface,

$$A_i = (0.2 \text{ m})(0.3 \text{ m}) = 0.06 \text{ m}^2$$

$$A_e = 2(0.02 \text{ m}^2 + 0.03 \text{ m}^2 + 0.06 \text{ m}^2) = 0.22 \text{ m}^2$$

Plug into the floating potential equation,

$$\frac{4v_0 A_i}{v_{e,th} A_e} = \frac{4(7.67 \text{ km/s})(0.06 \text{ m}^2)}{(213 \text{ km/s})(0.22 \text{ m}^2)} = 0.0392$$

$$V_{fl} = (0.0862 \text{ V}) \ln(0.0392) = (0.0862 \text{ V})(-3.24)$$

$$\boxed{V_{fl} \approx -0.28 \text{ V}}$$

Problem 4: Spacecraft Charging in GEO

Problem 5: Radio Wave Propagation

(i.)

To find the minimum radio frequency to communicate from ground to space, we would need the frequency that exceeds the largest plasma frequency ω_p in the atmosphere. From the IRI output, we find that to be,

$$\omega_p = 6.51 \text{ MHz}$$

(ii.)

To find the region where this frequency reflect, we need to find the region where radio waves are below this frequency. This would be below 225 km. Which would be the F2 region.

(iii.)

To find the region where at 2 MHz signal would reflect, we would treat that frequency as the plasma frequency to find in the IRI output. This would occur at about 97 km, which would be the D region.

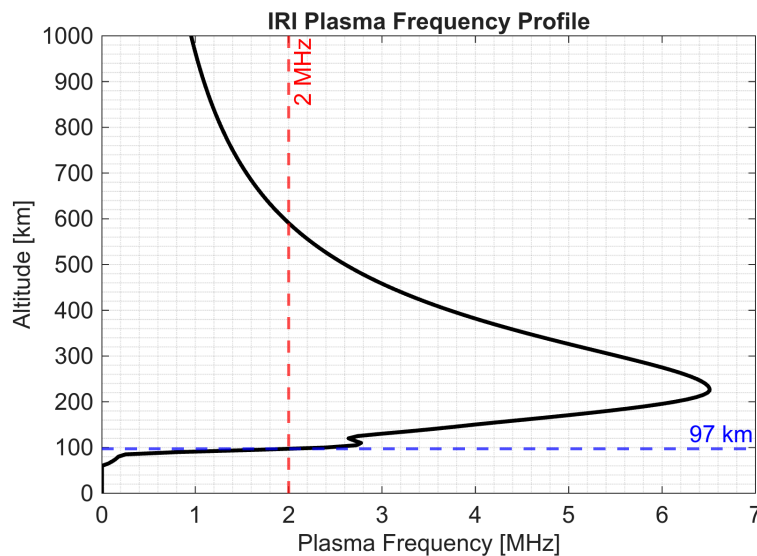


Figure 3: IRI Plasma Frequency

Problem 6: Radio Blackout

(i).

The reason that the spacecraft temporarily loses the ability to communicate with the ground is because during re-entry, there is an envelope of ionized atmosphere around the spacecraft, due to the re-entry processes' heat generated. This envelope of ionized atmosphere would have some plasma frequency that would cause radio signals to reflect, causing this radio blackout.

(ii).

If the module communicates with the ground at 2282.5 MHz, to find the electron density in the plasma sheath region, we use the equation,

$$f_c = 9 \times 10^{-3} \sqrt{n_{ec}}$$

rewrite for n_{ec} ,

$$n_{ec} = \left(\frac{f_c}{9 \times 10^{-3}} \right)^2$$

$$\boxed{n_{ec} = 6.432 \times 10^{22} \text{ m}^{-3}}$$

Problem 7: GPS Delays

(i).

We seek the path length delay through the ionosphere for 10 TECU using, $L1 = 1575.42$ MHz and $L2 = 2282.5$ MHz, we can use the equation,

$$\Delta l_{iono} = -\frac{40.3}{f^2} TEC$$

$$\Delta l_{ionoL1} = -1.62 \times 10^{-16} \text{ m}$$

$$\Delta l_{ionoL2} = -7.74 \times 10^{-17} \text{ m}$$

(ii).

We are able to use multiple frequencies because of the fact that ionospheric delay depends on frequencies. By modeling two frequencies provided by GPS satellites, we can then account for these ionospheric errors. By reducing this increased path due to the ionosphere, we can improve our pseudorange and have a better position estimate.

Problem 8: D Region Signal Attenuation

The electron-neutral collision frequency is found from the MSIS neutral density ($n_n = n_{N_2} + n_{O_2} + n_O$, cm^{-3}) and the IRI electron temperature,

$$\nu_{en} = 5.4 \times 10^{-10} n_n T_e^{1/2} \text{ s}^{-1}$$

The electron gyrofrequency for $|B_0| = 50,000 \text{ nT}$ is,

$$\omega_c = \frac{q_e B_0}{m_e} = 8.79 \times 10^6 \text{ rad/s} \quad (f_c = 1.40 \text{ MHz})$$

The differential absorption (with n_e in m^{-3}) for both the $+$ and $-$ solutions is,

$$\frac{dA}{dl} = 4.6 \times 10^{-5} \frac{n_e \nu}{\nu^2 + (\omega \pm \omega_c)^2} \text{ dB/m}$$

Integrating vertically through the IRI profile with $f = 10 \text{ MHz}$,

$$A_+ \approx 1.2 \text{ dB} \quad A_- \approx 2.1 \text{ dB}$$

Most of the absorption occurs between 65 and 120 km, peaking near 95 km where n_e increases rapidly while ν is still large. Neglecting B_0 ($\omega_c = 0$) gives 1.6 dB.

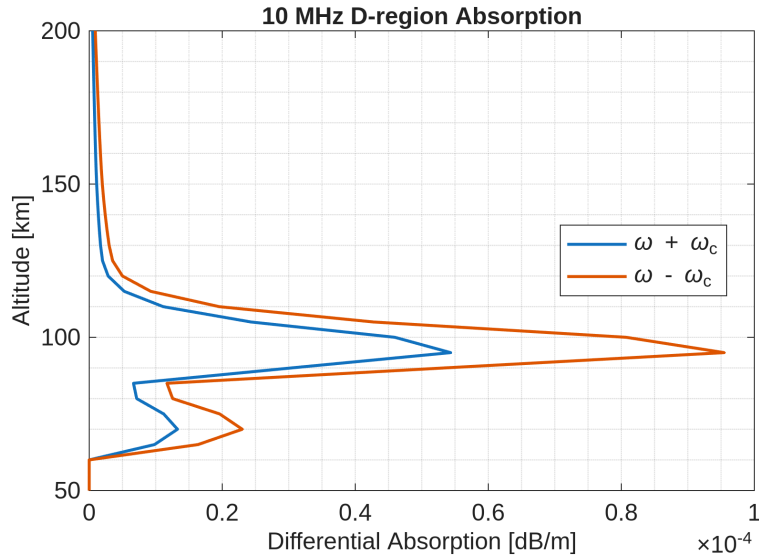


Figure 4: 10 MHz Differential Absorption

Problem 9: ICON Mission